AN INTELLIGENT PUBLIC TRANSPORT DELAY PREDICTION AND ROUTE OPTIMIZATION SYSTEM

Phase 1: Academic Literature Review, Research Gap Analysis & Formulated Objectives

LITERATURE REVIEW AND TECHNICAL RESEARCH REPORT

Student Level:	B.Tech Final Year (4th Year) Student
Academic Phase:	Phase 1 - Title, Literature Review & Objectives Formulation
Project Scope:	Public Transport Reliability, Delay Pattern Mining, and GIS Routing
Methodology:	Supervised Machine Learning, Spatial Clustering, and Explainable AI (XAI)
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University Status:	Undergraduate Final Year Project (Phase 1 Milestone)

Abstract:
This technical document presents an academic literature review of 10 foundational research papers in smart transit systems, travel time prediction, and route optimization. It systematically extracts critical research gaps, and translates these limitations into four high-priority, feasible project objectives for a year-long B.Tech development roadmap, backed by clear datasets, modeling strategies, and validation criteria.

1. Executive Summary

Public transportation systems are the backbone of modern smart cities. However, operational inefficiencies, primarily delays caused by traffic congestion, weather anomalies, and high passenger demand, degrade service reliability and user trust. This technical report provides a rigorous literature review of 10 key research papers to define the baseline for developing 'An Intelligent Public Transport Delay Prediction and Route Optimization System Using Machine Learning'.

By analyzing these papers, we identify critical limitations in current smart transit research. These include: (1) a severe lack of interpretability in high-performance predictive models (the 'black-box' problem), (2) a division between static scheduling optimizations and real-time operational disruptions, (3) vulnerability of models to missing or noisy sensor data, and (4) isolated stop-to-stop predictions that ignore downstream delay propagation. To bridge these gaps, this document formulates four high-priority, highly feasible objectives for the student's year-long B.Tech final project. These objectives focus on data fusion, explainable delay modeling, stop-level spatial clustering, and the creation of an interactive decision support system.

2. Project Background and Scope

The core project focuses on predicting public transit delays and optimizing routes to provide commuters and operators with reliable, actionable insights. Instead of simple 'late' alerts, the proposed system will explain delays (e.g., 'Route 204 is likely to be delayed by 14 minutes near Stop A between 8 AM and 9 AM due to morning peak traffic and historical patterns') and suggest optimal alternative routes.

The scope involves merging multiple open datasets including General Transit Feed Specification (GTFS) schedules, historical operational logs, meteorological records, traffic congestion scores, and calendar events. The project combines classification (delay probability), regression (delay duration in minutes), spatial clustering (stop-level hotspots), and an interactive planning dashboard (Streamlit/FastAPI) to deliver a complete smart transit analytics solution.

3. In-Depth Literature Review (Papers 1 - 3)

Paper 1: Predictive Modelling and Factor Analysis of Public Transport Delays (2025)

- **Authors & Source:** Y. Matseliukh, V. Lytvyn, Z. Hu, M. Bublyk | IJITCS, Vol. 17, No. 6
- **Methodology & Data:** Developed an eXtreme Gradient Boosting (XGBoost) model using a dataset of operational and meteorological logs from a city of ~1 million. Integrated Explainable AI (XAI) using SHapley Additive exPlanations (SHAP) for global and local factor analysis.
- **Key Findings:** The model explained 72% of delay variance (R2=0.72). Spatial context (stops, routes) and temporal cycles (hour, day, season) dominated predictions, accounting for over 52% of feature importance. Weather was secondary (45%), but its impact was highly contextual, being heavily amplified during peak operational hours but minimal otherwise.
- **Research Gaps:** 1. Fails to account for rare, unexpected events (e.g. accidents or road construction) not represented in historical logs. 2. Calculating SHAP values is computationally intensive, creating barriers for real-time edge integration.

Paper 2: Machine Learning-Based Route Optimization for Smart Transportation (2025)

- **Authors & Source:** A. M. Anson & Amirah | JOSAPEN, Vol. 3, No. 2
- **Methodology & Data:** Combined supervised learning (for travel time prediction) with Reinforcement Learning (for dynamic route selection). Leveraged GPS trajectories, traffic flow, weather reports, and OpenStreetMap (OSM) spatial data inside a simulated grid environment.
- **Key Findings:** The hybrid model outperformed static algorithms (Dijkstra and A*) in adapting to dynamic real-time disruptions like accidents, roadworks, and sudden weather changes.
- **Research Gaps:** 1. Grid-based simulation is highly simplified compared to continuous real-world street networks. 2. High computational complexity and long training cycles of reinforcement learning hinder scalability to city-level networks. 3. Completely ignores fixed transit timetables and bus schedule constraints.

Paper 3: Real-Time Passenger Train Delay Prediction (2022)

- **Authors & Source:** P. Lapamonpinyo, S. Derrible, F. Corman | IEEE OJITS, Vol. 3
- **Methodology & Data:** Compared Random Forest, GBM, and Multi-Layer Perceptron (MLP) under two architectures: Real-Time based Data-frame Structure (RT-DFS) using only current train coordinates, and Real-Time with Historical based Data-frame Structure (RWH-DFS) integrating previous train runs. Included ridership, census data, and weather.
- **Key Findings:** The MLP model using the RWH-DFS structure performed best. Adding exogenous features (ridership, weather, and population) improved R2 and MAE by 5.2% and 5.9%. The delay at the current station was the single most dominant predictor.
- **Research Gaps:** 1. The model is station-specific; scaling it system-wide requires building thousands of individual models. 2. Excludes mixed-use track conflicts, such as freight train interference. 3. Fails to handle dynamic data stream dropouts.

3. In-Depth Literature Review (Papers 4 - 7)

Paper 4: Enhancing Route Optimization in Road Transport Systems (2025)

- **Authors & Source:** N. El Karkouri et al. | Future Transportation, Vol. 5
- **Methodology & Data:** Compared ANNs, Random Forest, SVM, and Support Vector Regression (SVR) for trans-continental logistics optimization. Evaluated routes based on duration, cost (fuel, tolls), and safety (accident/infraction statistics) across the Dakhla-Paris corridor.
- **Key Findings:** Artificial Neural Networks (ANNs) achieved the highest accuracy in balancing duration, cost, and safety. Multi-criteria optimization successfully mapped safer and more efficient long-haul routes.
- **Research Gaps:** 1. Relies on static, seasonal climate averages rather than dynamic, real-time weather feeds. 2. Cross-border customs delays and regulatory waiting times are neglected. 3. Weighting of competing criteria (time vs. cost vs. safety) remains highly subjective.

Paper 5: Investigation of Bus Special Lane Performance & Intersection Delay (2022)

- **Authors & Source:** V. N. G. Gilani et al. | Journal of Optimization, Vol. 2022
- **Methodology & Data:** Evaluated dedicated bus lanes (BRT) using statistical analysis and optimized signalized intersection delays. Combined Geographic Information Systems (GIS) with metaheuristics and machine learning for traffic control optimization.
- **Key Findings:** Bus stop spacing of 200m to 400m in high-density urban areas is optimal. Intersections represent the main bottlenecks, and ML-optimized signal timings can reduce bus delays.
- **Research Gaps:** 1. Ignores passenger boarding dynamics, fare collection methods, and stop-level crowding. 2. Assumes homogeneous driver characteristics and constant vehicle acceleration. 3. Fails to model downstream gridlock spillback effects.

Paper 6: Public Transport Delay Prediction using Deep Learning (2026)

- **Authors & Source:** S. N. M. Rum, M. M. N. M. Yusoff, A. Mahdi | JARASET, Vol. 59, Issue 1
- **Methodology & Data:** Compared a Deep Neural Network (DNN) against standard regressors (Linear Regression, SVR, Decision Trees, Random Forest) utilizing public transport schedules and weather data.
- **Key Findings:** The DNN model achieved the lowest Mean Absolute Error (MAE) compared to other regression models, proving deep learning's ability to model complex tabular data.
- **Research Gaps:** 1. Complete lack of model interpretability (black-box). 2. Hyperparameter optimization was restricted to basic manual searches. 3. Fails to incorporate passenger loads or real-time traffic index variations.

Paper 7: Dynamic Traffic Flow Optimization Using Reinforcement Learning (2025)

- **Authors & Source:** V. N. Skoropad et al. | Sustainability, Vol. 17, No. 8
- **Methodology & Data:** Integrated Reinforcement Learning (DQN, PPO) for traffic signal control with predictive time-series models (LSTM, ARIMA, GNNs) using high-resolution loop detector data from Belgrade.
- **Key Findings:** The integrated system reduced intersection wait times by 33%, lowered emissions by 16%, and decreased local traffic accidents.
- **Research Gaps:** 1. The system does not simulate resilience to sudden, temporary traffic surges (e.g. mass concerts) or extreme weather. 2. Edge-based training of RL is highly resource-intensive. 3. Lacks prioritization rules for public transit vehicles.

3. In-Depth Literature Review (Papers 8 - 10)

Paper 8: Machine Learning for Next-Generation Intelligent Transportation (2022)

- **Authors & Source:** T. Yuan, W. D. R. Neto, C. E. Rothenberg, K. Obraczka, C. Barakat, T. Turletti | TETT, Vol. 33, No. 4
- **Methodology & Data:** A comprehensive survey of ML applications in ITS across infrastructure (sensing/comm), resource (edge/cloud), and orchestration layers, analyzing federated learning and cooperative driving.
- **Key Findings:** Outlined a major shift from traditional statistical analysis to deep learning and reinforcement learning. Highlighted the emergence of IoT-enabled vehicles and smart infrastructure.
- **Research Gaps:** 1. Lack of unified data standards across transit agencies prevents cross-city model deployment. 2. Severe data gaps and noisy GPS telemetry cause predictive models to fail. 3. Most studies are purely simulation-based, lacking real-world deployment details.

Paper 9: Soft GRU-Based Recurrent Neural Networks for Congestion Prediction (2023)

- **Authors & Source:** S. M. Abdullah et al. | Sustainability, Vol. 15, No. 7
- **Methodology & Data:** Developed a modified Gated Recurrent Unit (Soft GRU) featuring a soft gating mechanism to handle missing and noisy sensor inputs. Evaluated using GPS trajectories and traffic logs from India.
- **Key Findings:** The Soft GRU architecture outperformed standard LSTM/GRU networks, maintaining prediction accuracy and resilience even when up to 20% of sensor feeds were corrupted or missing.
- **Research Gaps:** 1. Recurrent neural networks have high computational overhead for continuous retraining on live telemetry. 2. Does not consider the feedback loop where showing congestion predictions alters driver routes and shifts traffic elsewhere. 3. Does not optimize transit routes directly.

Paper 10: AI-Driven Approach for Enhancing Sustainability in Public Transport (2024)

- **Authors & Source:** V. L. Vujadinovic et al. | Sustainability, Vol. 16, No. 17
- **Methodology & Data:** Applied Reinforcement Learning to optimize static public transport line elements (schedules, headway times, and buffer allocations) to mitigate headway deviations and prevent vehicle bunching in three Serbian cities.
- **Key Findings:** RL-based buffer time and schedule optimization significantly enhanced headway regularity, minimized passenger wait times, and improved transit energy efficiency.
- **Research Gaps:** 1. The optimization is static (done during scheduling) and cannot adapt dynamically in real-time to major road incidents. 2. Assumes constant passenger boarding rates, ignoring peak-hour surges. 3. Evaluated on mixed traffic bus lanes, and findings do not translate easily to grade-separated networks.

4. Synthesis of Key Research Gaps

Based on the literature review, the following key research gaps represent limitations that our B.Tech final-year project can address:

- **Gap A: Black-Box Predictions (Paper 1, 6):** Most high-accuracy models (XGBoost, DNN) fail to explain delay causes, leaving operators without actionable insights. Incorporating Explainable AI (XAI) is vital.
- **Gap B: Static Scheduling vs. Real-Time Incidents (Paper 7, 10):** Current optimization models are applied statically during schedule design. They fail to adapt dynamically in real-time when incidents, severe weather, or passenger surges occur.
- **Gap C: Data Gaps and Noise Sensitivity (Paper 8, 9):** Real-world GPS feeds are noisy and suffer from temporary dropouts. Standard models degrade significantly under these conditions, highlighting the need for robust preprocessing and imputation.
- **Gap D: Isolated Stop-to-Stop Predictions (Paper 3, 5):** Transit models often predict delays for a single isolated segment, failing to capture downstream delay propagation across the network.
- **Gap E: Lack of Actionable Planners Recommendations (Paper 1, 10):** Existing systems output prediction scores but fail to provide transit operators with operational suggestions (e.g. schedule changes, bus injection).

5. Formulated B.Tech Project Objectives

To address the identified research gaps while ensuring feasibility within a 1-year B.Tech project timeline, the following four prior objectives have been formulated:

Objective 1: Multimodal Data Fusion & Imputation Pipeline (Addresses Gaps B & C)

- Core Goal: Design a robust data ingestion pipeline that merges static schedules (GTFS), weather metrics, holiday logs, and traffic speed datasets.
- Imputation Strategy: Implement KNN Imputer or historical averages to handle missing/noisy GPS sensor data (up to 15% missingness).
- Dataset Feasibility: High. Involves open-source GTFS feeds (Google Transit/Transitland), historical weather (Kaggle/OpenWeather), and traffic indexes.

Objective 2: Explainable Machine Learning Model for Delay Prediction (Addresses Gap A)

- Core Goal: Train classification models (predicting delay probability) and regression models (predicting delay minutes).
- Modeling: Compare Random Forest, LightGBM, and XGBoost to determine the most accurate model.
- Explainability (XAI): Integrate SHAP (Shapley Additive exPlanations) to explain predictions globally (overall route bottlenecks) and locally (why a specific bus is late).
- Feasibility: High. Well-supported Python libraries (scikit-learn, xgboost, shap) are available.

Objective 3: Stop-Level Spatial Clustering & Downstream Propagation (Addresses Gap D)

- Core Goal: Perform spatial clustering (using DBSCAN or K-Means) on coordinate data to locate stop-level delay hotspots.
- Propagation Modeling: Calculate stop-level average dwell times and use sequence features to model how delays propagate to downstream segments.
- Feasibility: High. Leverages GIS libraries (GeoPandas, Folium) and basic clustering algorithms.

Objective 4: Interactive Decision Support Dashboard with Actionable Insights (Addresses Gaps

- Core Goal: Build a dual-user dashboard (Passenger View: predictions and suggestions; Admin View: route metrics and recommendations).
- Operator Recommendations: Generate rule-based recommendations (e.g. 'Route 204: Delay probability at Stop A exceeds 75% between 8-9 AM. Action: Inject 5 min buffer time or add one extra vehicle').
- Commuter Routing: Implement Dijkstra/A* pathfinding on OpenStreetMap (OSM) weighted by predicted delays.
- Feasibility: High. Python + Streamlit + Plotly + Folium provides a lightweight, highly visual web application stack.